

Assessing the Disturbance from Overcrowding in Outdoor Nature Experiences

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Figure 1: Rök. A portable device for assessing overcrowding in nature experiences.

ABSTRACT

We present a new device designed for field studies which have the goal of assessing the perception of overcrowding in nature experiences, with a special focus on areas subjected to mass tourism (pre COVID-19). The design of the device resulted from an interdisciplinary approach attempting to mix valuable techniques from User Experience exploratory research, with the typical way of conducting overcrowding investigations in outdoors and wild areas.

*Both authors contributed equally to this research.

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We began by reframing both nature and overcrowding as *experiences* and defining overcrowding as a disturbance to the nature experience. This led to a new idea for a more beneficial investigation of visitor's perception, which is moment-to-moment and in context. The device supports recording someone's subjective perception of overcrowding in space and time, with minimal task load or distraction from the nature experience. The streamlined design affords only two buttons with competing functionalities: one to signal "there are too many people at the park" and another for "the park is too empty". The device continuously records time, location, and press-down button events, storing everything locally and anonymously, within the boundaries of European GDPR. We applied our method to a field study at the Þingvellir National Park in Iceland, both a protected area and a UNESCO World Heritage. We then analyzed the collected data and visualized where and when the visitors reported overcrowding. Initial results indicate that by complementing traditional questionnaires with moment-to-moment self-reporting, we could successfully estimate the disturbance from

overcrowding over time and place, thus producing a deeper insight into the quality of the experience than questionnaires alone. Our results also speak of some intrinsic qualities of the outdoor infrastructure. The outcome has the potential to make park managers, designers, landscape architects and rangers more capable of understanding the complex interrelation between infrastructure and visitor flow, thus contributing towards the goals of a long-term sustainable management of the area.

CCS CONCEPTS

• **Human-centered computing** → **Contextual design**.

KEYWORDS

overcrowding, nature experience, spatio-temporal mapping, in the wild method

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1 INTRODUCTION

In recent years, Þingvellir National Park saw a large increase in the number of visitors due to mass tourism in Iceland post-financial crisis. In a matter of a few years the number of people visiting the area nearly doubled [8]. The change happened so abruptly to leave the infrastructure largely unprepared against a much larger volume of visitors. As a consequence, the national park developed intense overcrowding which according to locals ruined the authentic experience of the area. Before COVID-19 - when the ease of international traveling allowed for mass tourism - it was natural to question how much of the authentic character was left for visitors to experience during summer season.

One of the most authentic experience of nature is one of solitude, where someone can have an intimate encounter with nature itself. On the other hand, we all have experienced a naturalistic area where we feel surrounded by "more people than nature". In highly popular outdoor locations the perception of overcrowding diminishes the authentic character and quality of the experience [2]. This experiential degradation indicates room for improvement in the preservation of the area, especially in light of a long-term sustainable strategy [13]. The dynamic interplay between the visitor flow, the characteristics of the infrastructure and the nature experience is so complex which, we believe, is hard to tackle without technological assistance. The risk is a struggle to reach a consensus between decision makers on the best strategies for a sustainable preservation of the area.

When practitioners and researchers attempt to measure overcrowding in the outdoors, they typically rely on questionnaires which are handed to visitors at the end of their experience [7]. A questionnaire-based study can be effective but also has the limitation of probing someone's impressions after the fact rather than during the experience. We argue that this approach is not ideal to

assess a perceptual phenomenon like overcrowding which is contingent on time, place and the contextual background created by the presence and behavior of other people, especially if we are looking to get a global view of the phenomenon for a large area. To advance the state of the art, we took an interdisciplinary approach that lead us to the design of a new device: Rök. By using this custom made device, we were able to define a methodology for an experimental field study that complements the typical questionnaire investigation with a form of lightweight ethnography.

In particular we were keen on contextual enquiry, an effective tool in the practice of User Experience exploratory research. We wanted to retain the ability to observe a person in context moment-to-moment during the experience, rather than after the fact. However we soon concluded it was impractical to follow a visitor through the park, potentially producing a stronger disturbance than overcrowding itself. Therefore we reframed the contextual observation as *contextual self-reporting*. Reframing the problem created room to envision a portable device that visitors carry on as they walk through the park, continuously collecting data about where and when they felt there were too many people at the park or too few. Once we could collect data with this degree of granularity, we were able to design a series of detailed visualizations of the overcrowding phenomenon over space and time.

Within a larger picture, we see our work as contributing to the preservation of an authentic nature experience. Starting from an idea of authenticity akin to the concept of *existential authenticity* (see [19] for a more comprehensive dissertation), we approached the problem through *via negativa* [18] and attempted to gain knowledge about what interferes with authenticity. With this information at hand, we think that park managers are better equipped to decide on strategies and infrastructural interventions that effectively can reduce overcrowding, maintain a quality nature experience without limiting access, and contribute to the long-term sustainable management of the area.

2 RELATED WORK

Overcrowding is a problem that afflicts many different tourist locations. Studies have shown a correlation between overcrowding and tourist dissatisfaction [3, 12, 17]. Being surrounded by too many people can ruin the experience [2], which in turn diminishes the probability of choosing an overcrowded destination compared to another offering a more authentic experience [1]. Research on this topic has led to the definition of the term *psychological carrying capacity* [16] as the impact that people have on other people while visiting the area [10, 11, 15]. Some type of visitors, for examples casual tourists vacationing away from metropolitan areas, might find acceptable if not reassuring to be visiting an area while surrounded by other people. On the other hand, locals or hikers may look for a more solitary experience which doesn't include masses of people in colourful clothes. The range of categories of visitors can be arranged along a purism scale [16], based on their expectation about the authenticity of the nature experience. When the management of the area responds to the expectations of only a part of the purism scale, for example casual tourist, the planning of new infrastructures tends to polarize in one direction only, with an increased risk of deterioration of the area in the long run [16].

Since overcrowding is a subjective perception, researchers often conduct surveys to estimate the intensity of the phenomenon as it appears to people. Typically they are conducted via questionnaire, either self-administered or with assistance. Questionnaires have advantages over interviews [6, 9]: usually cheaper and faster, have a broader reach, compel more honest responses and are less prone to the interviewer bias. On the other hand questions might get misinterpreted and little room is left for a person to express a more nuanced opinion.

Questionnaire surveys have been employed in many different areas, including studies of social carrying capacity for touristic locations, which share a similar focus to ours. For example, Gonson et al. [5] employed interviews and questionnaires for assessing the social carrying capacity of recreational coastal areas [5]. They collected data on the perception of crowding and potential causation factors. They concluded a possible correlation between the sensation of crowding and spatial distribution of tourist. A questionnaire survey has also been conducted by [7], this time to estimate how much the overcrowding was a disturbance to the experience of a certain location. They then made a model to correlate the number of people entering the area with the intensity of the overcrowding and used it to propose a threshold to the maximum number of visitors allowed in the area before the experience degrades.

While this kind of survey produces some useful information about the cognitive state of a person, under certain conditions it may suffer *retrospective bias*. One's memory and perspective can make such reporting unreliable [6]. There is mounting evidence for the impact of retrospective bias while reporting on past emotions. Robinson and Clore [14] does a great job of summarizing the existing literature on this topic. Concerning our own study, a precise assessment of overcrowding could be negatively impacted by (i) poor posterior assessment the feeling of overcrowding (ii) greater tendency to report a feeling about the entire visit, instead of the place to place feelings (iii) downplayed reporting of strong feelings, based on the expectation of experiencing crowds (e.g. considering the location a "tourist trap" and expecting to having to deal with it).

More recent research "in the wild" has focused on transitioning from questionnaires into a form of moment-to-moment data collection. For example, Georges et al. [4] explain how they created a device that triangulates GPS coordinates, and samples bio-metric data to create emotional maps. The data points were then plotted onto maps that highlight the most exciting park areas. The data was substantiated with questionnaires, to investigate the correlation between physiological recordings and excitement self-assessment. Although they successfully acquired high quality data, the cumbersome device can subtract from the experience of the place and bio-metric signals still require an interpretation to be associated with emotions. It's an experimental method that trades off accuracy for costs: each trial lasted 4 hours because of the overhead caused by managing the device.

3 BACKGROUND RESEARCH

Overcrowding affects negatively the nature experience at Þingvellir National Park in Iceland. It creates congestion that makes the flow of visitors more problematic in certain areas and creates a disturbance to the overall experience. In general, overcrowding elicits

the perception of a "tourists trap" which demotes the naturalistic, cultural and historical significance of the park. Overcrowding interferes experientially with the authentic character of the area, both for locals and newly arriving visitors. It becomes evident by having casual conversations with the local Icelanders, keepers of the most truthful character of Þingvellir. Upon informal interviewing with the locals, we found that many stopped going to Þingvellir during high season because of how much the experience felt degraded.

By visiting the park and interviewing the rangers we found that overcrowding was happening in certain parts of the trail, at certain hours and during certain days. The actual conditions were uncertain but they had some ideas connected to the arrival time of busses and the arrival dates of cruises which greatly increase the visitor pressure at the park. The planning for better infrastructure is currently underway but the classic problem remains of having to balance different voices from different stakeholders, that are often perceived as having a vested interest or not being in touch with the economical reality at hand.

Visitors arrive at Þingvellir either by car or by bus, and the ones arriving by bus do so by following the much stricter schedule of bus arrivals. Most of the busses arrive in the morning between 10:00 and 12:00 and when large cruises arrive at the harbour in Reykjavik, the number of morning busses nearly doubles. Visitors arriving by bus are more likely to rush through since they are bound to the bus schedule which gives them only 20 to 45 minutes to explore the area. Conversely, the people arriving by car tend to stay for more than 1 hour to enjoy the area without time pressure.

The natural conformation of Þingvellir and the position of the main parking lot don't leave much opportunity to reroute the visitors in alternative ways as of now. However, after an interview with the rangers, it became clear that they didn't have much data about the visitors flow in general, therefore most of the discussions on immediate interventions, such as changes to the signage system, tended to get diluted by a myriad of cost-benefit speculations.

4 CONCEPT AND IDEATION OF THE DEVICE

The idea of making a rugged data collection device was inspired by the Nordic outdoors, often harsh on the fragile consumer tech, and our findings during the initial background research at the park. We knew that some visitors were rushing through and the bulk of the recruiting would have been entirely on us. We concluded that handing an object rather than requesting to install an app would be more resilient against a number of unexpected scenarios that we didn't have control over.

A physical object can be handed to a tourist without extra preparatory steps, like downloading or installing anything. We also believed that by asking people to install an app on a personal device like a phone, we could have elicited the perception of a vested interest in privacy breach and therefore demotivate some people from participating in the research. An app running on a phone, in an outdoor context, tends to be somewhat fragile against unexpected events. The person might accidentally lock the phone or switch to another app to, say, take a photo or even receive notifications from other apps running in the background. There are so many attention grabbers on a smartphone that they will likely interfere with the self-reporting activity, which is at the core of the research

we conducted. Moreover even during summer, Icelandic outdoors can be chilly and windy and some people prefer to wear gloves that are typically not smartphone friendly. On the other hand, to make a prototype device with Arduino and 3D printing is quick and inexpensive. A physical object provides a dedicated usage and a tangible functionality which is immediately accessible without much thinking, looking or fiddling. By making a non-consumer device, custom made for research purposes, we guarantee both effective and perceived anonymity and privacy protection at face value.

The device strives for a streamlined design and functionality. It's meant to fit into someone's hand, it's portable and heavy enough to remind of its presence. The functionality is minimal, focused and immediate, relieving the person from any implied expectation of task performance. It is intended to be resilient against arctic and sub-arctic outdoors conditions, often unpredictable. It is compact, self-contained and sturdy to resist shocks and a range of rough atmospheric circumstances typical of Icelandic weather. Like a common stone laying on the ground, it's available to be picked up at the entrance and then dropped before leaving the park. It's silently self-contained and presenting only two available operations for self-reporting. The device is called Rök (figure 1).

Two buttons, upper and lower, are on the top surface of the device. The person can press the upper button whenever they feel "there are too many people at the park", while pressing the lower button when they feel "the park is too empty". The available operations are only two in order to minimize cognitive load and ambiguity, with the added value of not distracting from the nature experience and be more resilient against unexpected use cases in the outdoors.

The device is portable, rugged, water-resistant and powered by a rechargeable battery. It's "alive" in the sense that continuously records its time and location as well as every button press-downs (figure 2). An internal memory stores the information locally and anonymously, within the boundaries of European GDPR.

5 OVERVIEW OF THE STUDY

We combined elements of UX Exploratory Research with the traditional questionnaire typical of tourism field studies. The study

is designed around the capabilities of the portable device which complements the results from the questionnaire. The questionnaire is divided into two parts, one filled out before starting the visit and the other one at the end of the visit. The first part focuses on demographics, while the second one investigates the overall visiting experience. We recruited participants upon arrival at Þingvellir and filled the first questionnaire part out with them, after providing instructions on how to use the portable device. The second part of the questionnaire was provided upon the person returning the device after finishing their tour at the park. Every questionnaire was timestamped with the initial time of recruitment and associated with the device handed to the participant. The goal was to gain more contextualized information about the perception of overcrowding by complementing the results produced via questionnaires with a spatio-temporal mapping of the phenomenon.

The field study lasted 9 days between July and August 2019 (high season at Þingvellir National Park), each day covering the hours from 10:00 to 16:30. Despite the practical challenges of trying to engage rushing tourists in a research effort, we were able to on-board a decent total amount of 66 participants. Figure 3 shows the workflow of the recruiting process for one participant. You can consider the figure as a *user journey for the researcher*, which is loosely based on UML activity diagrams with additional storyboard vignettes. We propose a diagram like this to facilitate the reproducibility of the study. It was important to provide simple and clear instructions about the device functionality and always recite the same script verbatim in order to increase the validity of the collected data:

- "You press the **upper button** whenever you feel there are too many people in the park."
- "Instead, you press the **lower button** whenever you feel the park is too empty."

The data collected was completely anonymous and the use of a custom-made device assured that the data was not taken from a personal device such as a smartphone. The data was stored locally in the device and therefore, in accordance with the European GDPR, it was enough to give every participant a paper explaining the purpose of the research, the lack of risks involved and their agreement in letting us work with the anonymous data collected.



Figure 2: Example of data gathered from a single participant by the device.

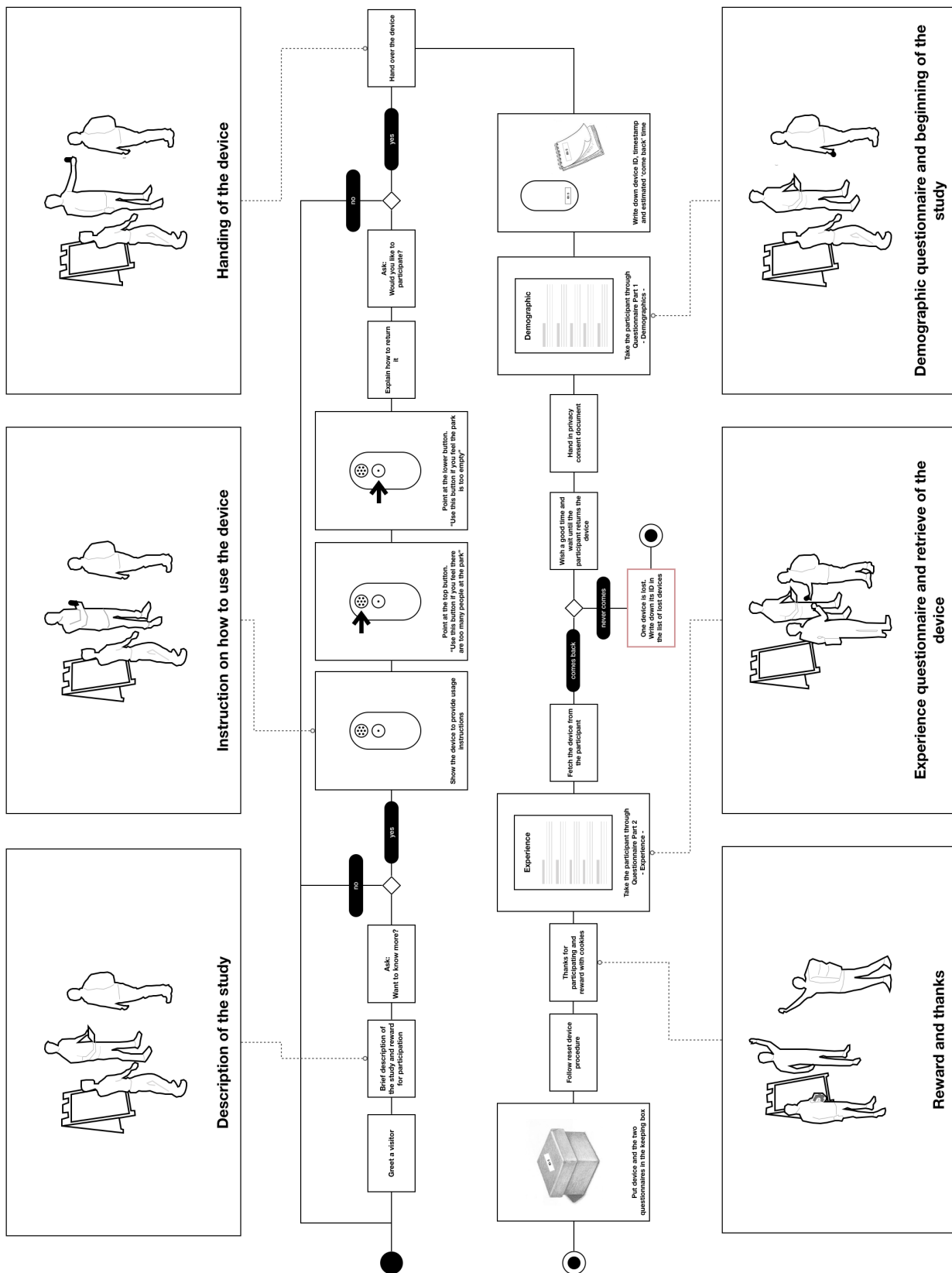


Figure 3: Flow diagram of the recruitment process.

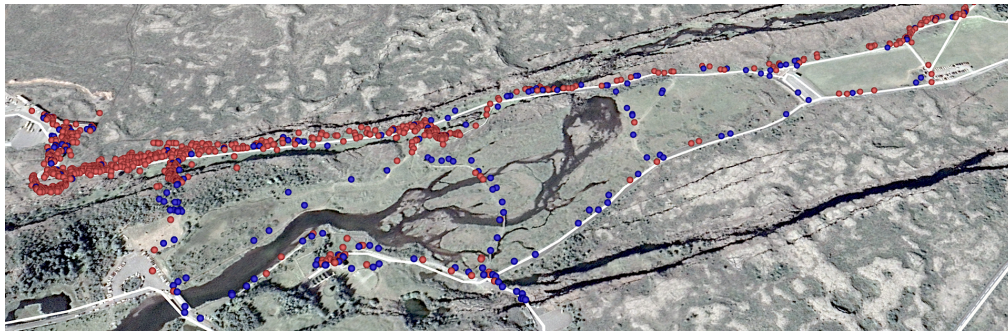


Figure 4: Mapping of the perception of crowding at Þingvellir National Park. A cumulative point cloud of the button press events generated by the participants between 10:00 and 16:30. The red dots are crowded presses while blue ones are isolated presses.

6 DATA PROCESSING AND VALIDATION

After the field study, we started focusing on ways to clearly map the data. We realized that getting the client for *Google Earth* we could easily overlay the data over the park area.

Not all of the data was usable though. Upon examination of the device output files, we found that some of the data-points were corrupted. Either the participant misused the device by switching the meaning of the two buttons or by overusing them (perhaps usage instructions were not delivered correctly) or the device was malfunctioning altogether (hardware issues that were fixed after the first couple of days). Some participant had clearly swapped the meaning of the two buttons as it was evident by comparing their data set with the one of others recorded during the same time period. Finally, after computing the frequency of press-down events per hour, we noticed some outliers that indicated a possible additional device malfunction. Some participants appeared to be pressing both buttons more than 100 times per hour, which was way higher than the others (but still possible).

After the filtering faulty data we were left with 49 valid data-sets, 32 from people arriving by car and 17 from people arriving by bus (from both regular bus tours and cruises). Any further processing was performed on an aggregate master data-set of valid data. We then proceeded to visualize the press-down events as a point cloud: a contextualized mapping of the visitor self-reported perceptions overlaid on a top down of the national park (Figure 4). Achieving this image was already a valuable result from the field study. A higher concentration of points means a more intense perception of either overcrowding or isolation.

Moreover, thanks to continuous GPS triangulation we could trace the walking path of every participant. Figure 2 shows a partial view of the path walked by one person. By taking a first look at the data-sets we could quickly conclude that they were plausible given our preliminary interviews with the park rangers. However, in order to be more confident about our results we deepened the analysis further.

7 ANALYSIS

How can we ascertain that the density of dots on the point cloud image is correlated with the visitors perception of overcrowding?

The intensity of the perception is proportional to the number of people ([7]) therefore we decided to investigate whether the number of the upper button press-downs would correlate with the number of people at the park. Our hypothesis was that the "intensity of self-reported sensation of overcrowding" was indeed related to the number of press-down events that were recorded at a given time. Our assumption was that the device was successfully designed to afford a person to report their sensations.

The park managers gave us access to gate counter data that tracks the number of people at the park on a 15 minutes interval. By using the data from the gate counter we plotted a graph of the average number of people at the park over the duration of our field study. We then overlaid a graph of the number of press-down events with the same 15 minutes time interval. The result is shown on figure 5 together with a linear regression showing a strong positive correlation with a p-value < 0.05.

To capture at first glance the intensity of overcrowding in various places of the park, we generated a height-map of the cumulative press-down events from the point cloud of figure 4. We sampled the entire area into a 5 meters grid and for every cell we calculated the accumulation of the the press-down events that fell within it. The result is shown in figure 6. The peaks show where the intensity of the overcrowding has been reported as cumulatively higher across the entire day. To highlight the peaks at first glance, we produced a heat-map from the height-map data (part below of figure 6). The figure is densely informative and shows what areas are the most affected by overcrowding and how they fare in comparison to one another. Once presented to the rangers, the visualization appeared plausible and quite telling of what parts of the infrastructure should be brought into a deeper scrutiny for future planning.

Figure 6 is designed to provide a global cumulative overview of the intensity of crowding but it's a static image and cannot capture those features of the phenomenon that evolve over time. To overcome this limitation, we went on to design an interactive spatio-temporal visualization of the overcrowding, with the addition of a controllable timeline. The idea is to allow the exploration of the temporal dimension and to illustrate how it correlates with relevant information regarding the arrival time of the visitors at the park. A screen shot of the interactive panel is shown in figure 7 and a video is available in the supplemental materials. Moving

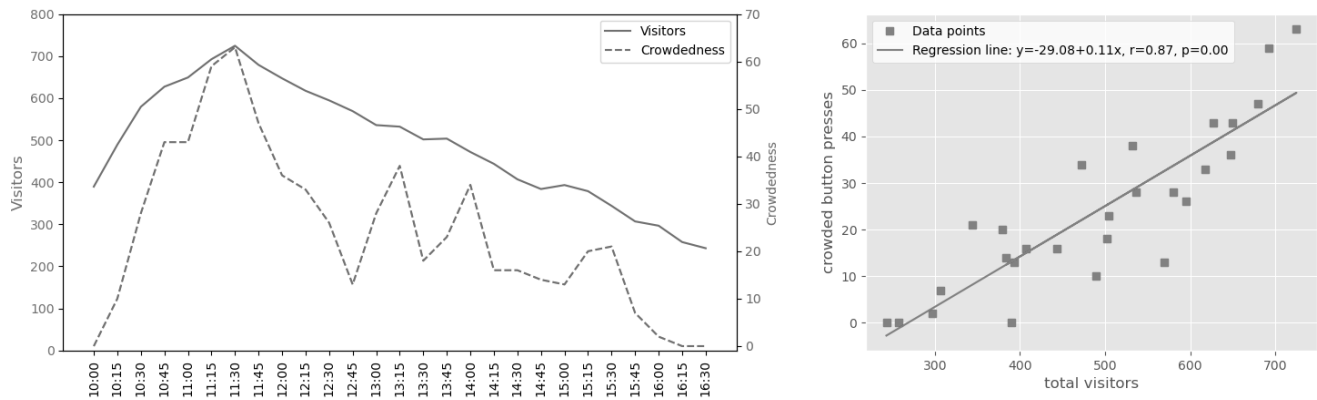


Figure 5: (Left) A graph showing profile similarities between the number of people at the park (solid line) and the intensity of the reported overcrowding (dashed line). The vertical scale has been adjusted to visually highlight this correspondence. **(Right)** A regression analysis confirming how the two curves are strongly correlated, with $p\text{-value} < 0.05$.

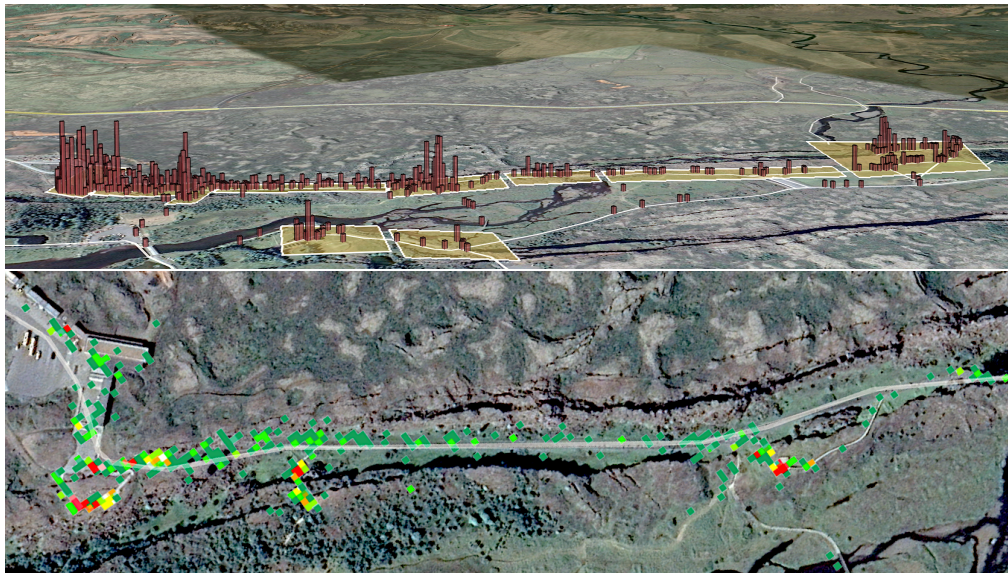


Figure 6: (Above) Height-map of the overcrowding showing the cumulative number of upper button press-down events on a 5m grid. **(Below)** Overcrowding heat-map. Warmer areas, tending to red, indicate reported intense overcrowding.

the slider at the bottom will animate the height-map, showing how the intensity of overcrowding evolves during the day. This interactive visualization could become a valuable discussion tool during stakeholders meetings.

8 CONCLUSIONS

This paper has described a novel way to assess the perception of overcrowding at a national park, providing rangers, managers and architects with a baseline for coming up with, discussing and eventually deciding on strategies towards sustainable tourism and preservation of authenticity. Rather than passively observing the visitors, they can be recruited for self-reporting with the right kind of device. While the overall experience and data analysis provide strong support for this approach, a couple of limitations of the

method should be considered. First, the devices have to be retrieved back from the visitors, which can be logistically tricky, especially with multiple exits from the area. This can limit the amount and rate of recruitment. Second, while valuable insights can be gained from the button presses, especially when aggregated over a large number of visitors, there is limited context for each press. It is impossible to say precisely what caused the press. The device could possibly be extended to record more contextual information such as body orientation, proximity to other devices, physiological signals or perhaps offer verbal commentary.

In sum, the contribution of this case study lies in its interdisciplinary UX approach to assessing overcrowding in nature experiences, where traditional questionnaires have typically been employed without any possibility for spatio-temporal mapping.

Thingvellir National Park

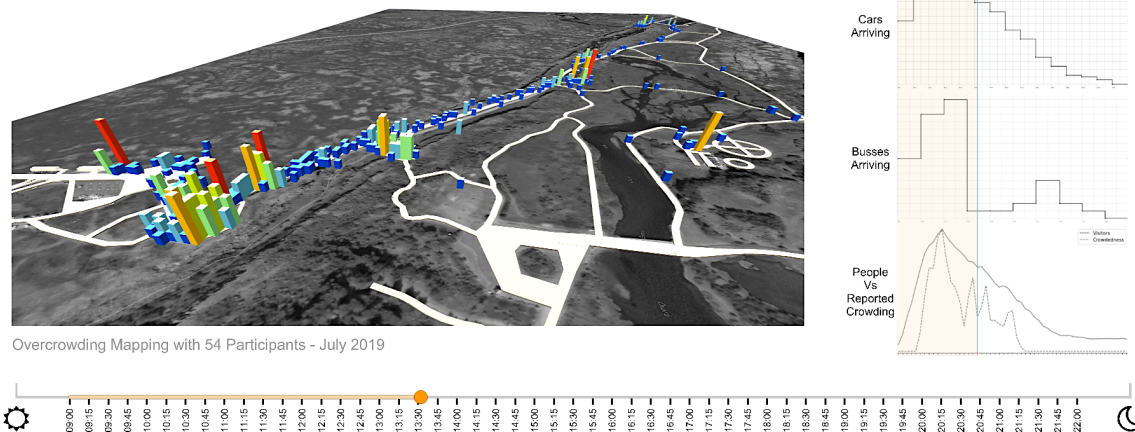


Figure 7: Interactive spatio-temporal visualization of overcrowding at Þingvellir National Park. The panel provides additional information that affects the number of people at the park, such as the arrival times of busses and cars.

The method is described using a user journey for reproducibility, and all data can be made available upon request. Beyond the current study, the use of the device could be extended to other types of perceptual phenomena. For example, it could be used to assess visual disturbance caused by new infrastructures in nature, or assessing what urban areas feel more exposed to COVID-19 hazards based on people's perception of excessive crowding or enclosed spaces.

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